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WO-300-05 — Replace VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 (4"-P300-P-1005/B13)

Area	A300 Exchangers and drum	**Task**	VR Valve replacement in kind
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	E-101 — E-101 shell-and-tube exchanger 36" ID × 192" — E 111'-6" N -25'-0" EL 106'-6"		
Equipment	E-101	**Lines**	4"-P300-P-1005
Isometrics	ISO-P-1005 (owner's drawings, attached)		
Permits	General work permit		

Reason

Control valve trim worn; replace the valve in kind.

Scope of work

1. Isolate per the isolation list, drain and vent 4"-P300-P-1005 between the isolations.
2. Unbolt and remove VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 (4"-P300-P-1005/B13) at E 33'-6" N -15'-7-15/16" EL 121'-0".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 of the same size, class and face-to-face (CONTROL4-P300), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 4"-P300-P-1005/B13, 4"-P300-P-1005/B15, 4"-P300-P-1005/B12

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1005	4"-P300-P-1005	4"-P300-P-1005	4	0 error(s), 5 warning(s) — see below

ISO-P-1005 (PDF): ../isometrics-imported/ISO-P-1005.pdf — the sheets that hold the work follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1005	1	warning	weld 4"-P300-P-1005/W11 (E 109'-11/16" N -4'-0" EL 116'-6", sheet 1) has no label
ISO-P-1005	3	warning	W18 (4"-P300-P-1005/W18, sheet 3) is labelled again on sheet 2; it is 223.3 in from the nearest break
ISO-P-1005	3	warning	weld 4"-P300-P-1005/W19 (E 34'-11/16" N -2'-0" EL 116'-6", sheet 3) has no label
ISO-P-1005	4	warning	weld 4"-P300-P-1005/W21 (E 33'-6" N -2'-0" EL 117'-11 5/16", sheet 4) has no label
ISO-P-1005	2	warning	branch weld 3/4"-P300-I-7005/W1 (E 109'-0" N -2'-0" EL 116'-9 5/16", sheet 2) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
E-101	equipment	E-101 shell-and-tube exchanger 36" ID × 192"		E 111'-6" N -25'-0" EL 106'-6"
4"-P300-P-1005/B13	item	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300	4"-P300-P-1005	E 33'-6" N -15'-7-15/16" EL 121'-0"
4"-P300-P-1005/B15	item	GASKET 4" P300 COMPRESSED SHEET	4"-P300-P-1005	E 33'-6" N -14'-11-11/16" EL 121'-0"
4"-P300-P-1005/B12	item	GASKET 4" P300 COMPRESSED SHEET	4"-P300-P-1005	E 33'-6" N -16'-4-1/8" EL 121'-0"
4"-P300-P-1005/B16	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1005	E 33'-6" N -14'-4" EL 121'-0"
4"-P300-P-1005/B10	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1005	E 33'-6" N -16'-11-13/16" EL 121'-0"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P300-P-1005/B16	4"-P300-P-1005	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1005/B16)	E 33'-6" N -14'-4" EL 121'-0"		
2	blind	4"-P300-P-1005/B13	4"-P300-P-1005	Fit a line blind between 4"-P300-P-1005/B16 and VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 (4"-P300-P-1005/B13), work side	E 33'-6" N -16'-4-1/16" EL 121'-0"		
3	valve	4"-P300-P-1005/B10	4"-P300-P-1005	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1005/B10)	E 33'-6" N -16'-11-13/16" EL 121'-0"		
4	blind	4"-P300-P-1005/B13	4"-P300-P-1005	Fit a line blind between 4"-P300-P-1005/B10 and VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300 (4"-P300-P-1005/B13), work side	E 33'-6" N -16'-4-1/16" EL 121'-0"		

Parts

Description	Qty	Replaces
VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300	1	4"-P300-P-1005/B13
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1005/B15
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1005/B17
GASKET 4" P300 COMPRESSED SHEET	1	4"-P300-P-1005/B12
STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	1	4"-P300-P-1005/B14

Inspections

Id	Inspection	Criteria	Basis
WO-300-05-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
4"-P300-P-1005/B15	4 × 1" studs (4"-P300-P-1005/B17)	35,343	3	
4"-P300-P-1005/B12	4 × 1" studs (4"-P300-P-1005/B14)	35,343	3	

Test — Service leak test

Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

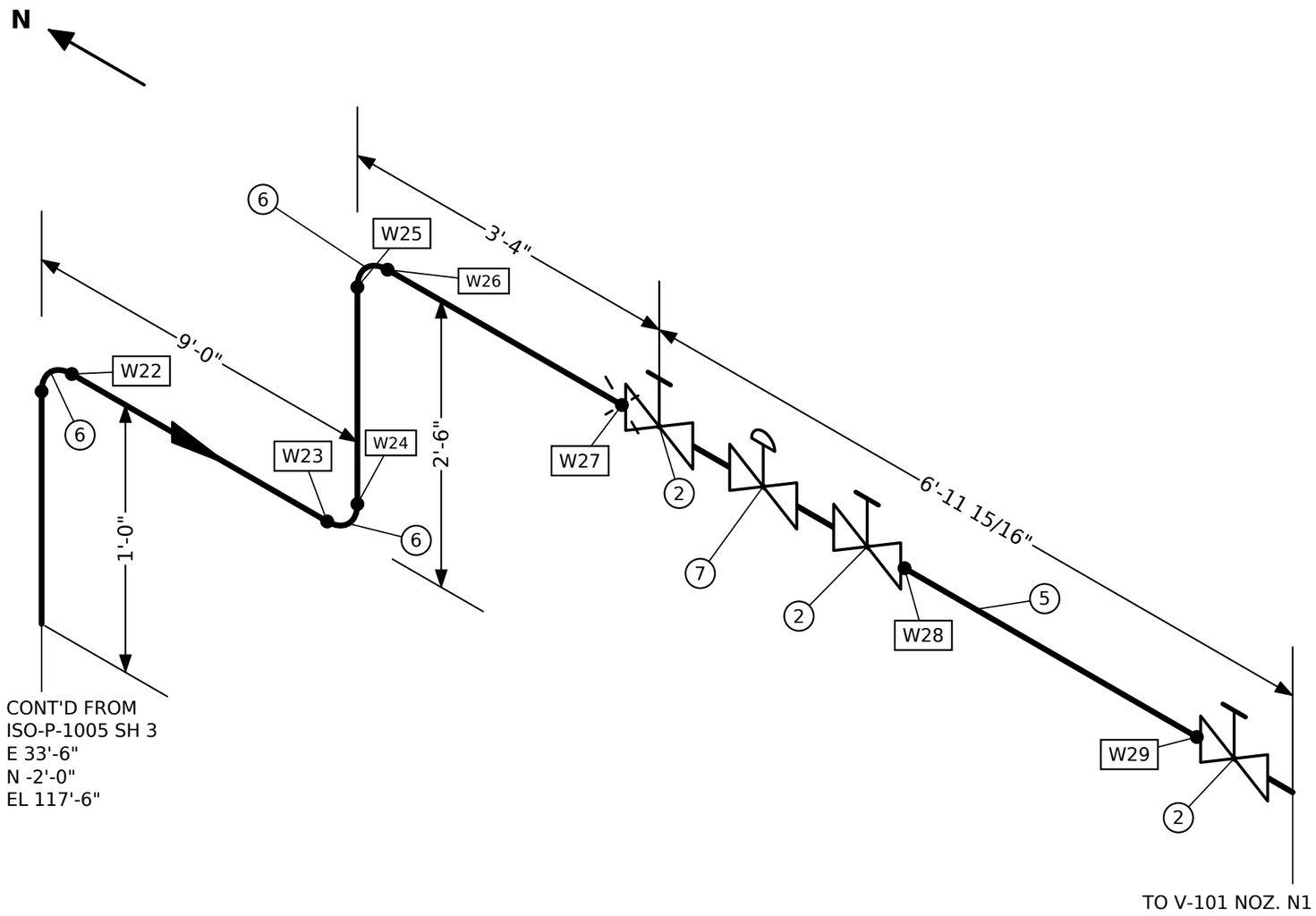
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 4"-P300-P-1005/B13 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	CONTROL4-P300	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
4"-P300-P-1005/B13	affected	ISO-P-1005	4	
4"-P300-P-1005/B15	affected	ISO-P-1005	4	
4"-P300-P-1005/B12	affected	ISO-P-1005	4	
4"-P300-P-1005/B16	isolation: valve	ISO-P-1005	4	
4"-P300-P-1005/B13	isolation: blind	ISO-P-1005	4	
4"-P300-P-1005/B10	isolation: valve	ISO-P-1005	4	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	6	4"	GASKET 4" P300 COMPRESSED SHEET
2	3	4"	VALVE, GATE, WEDGE, OS&Y 4" P300
3	6	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300
4	3	4"	FLANGE WELD-NECK 4" P300 RAISED FACE
5	12'-10 13/16"	4"	PIPE 4" P300 BEVELLED ENDS
6	3	4"	ELBOW 90° R = 5/3 OD 4" P300
7	1	4"	VALVE, CONTROL, GLOBE WITH DIAPHRAGM ACTUATOR 4" P300

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1005	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1005	4 OF 4	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

Notices

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